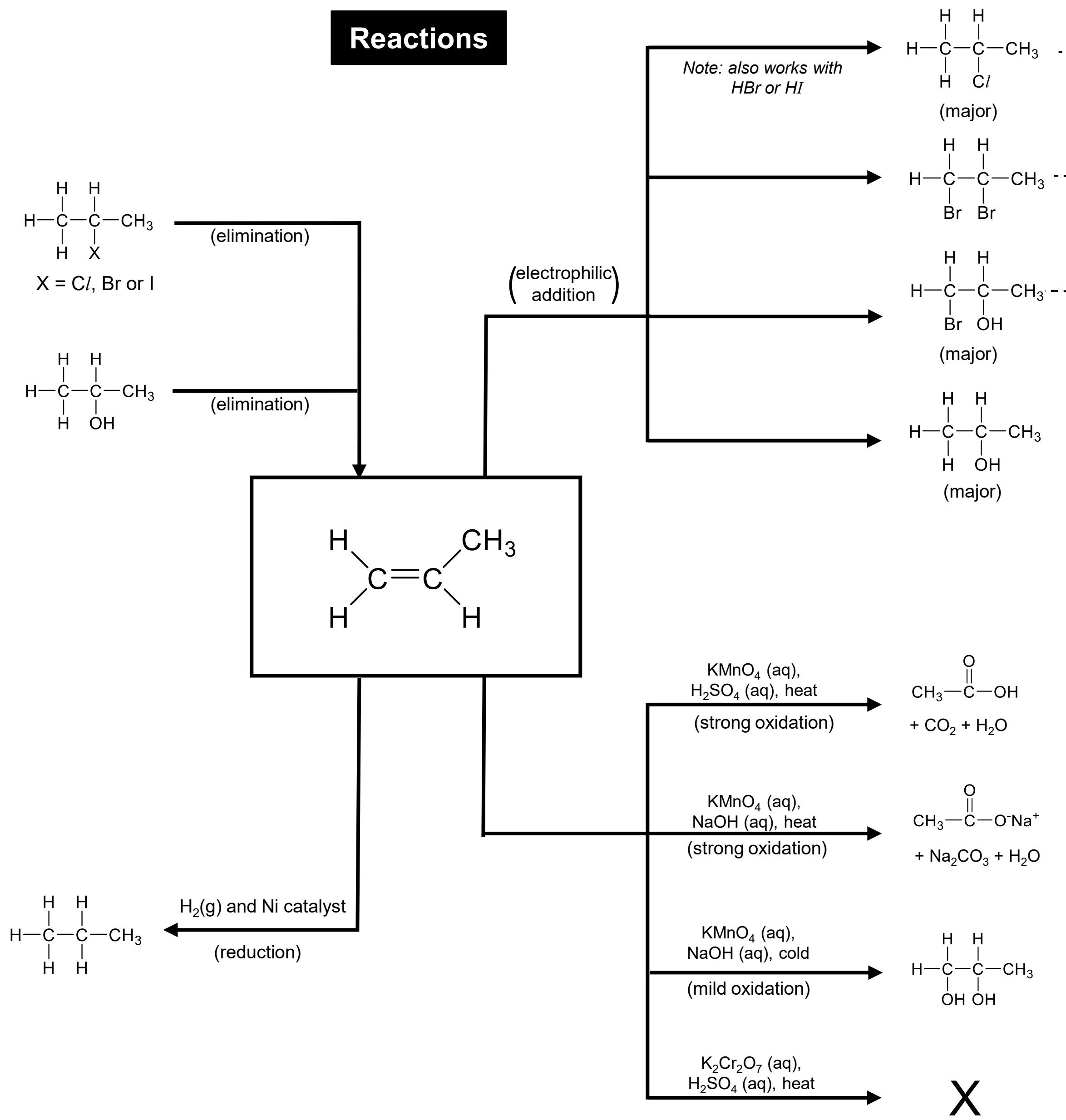
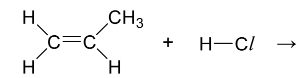


Mechanism

Name:

Step 1:

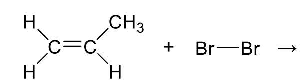


Step 2:

Mechanism

Name:

Step 1:



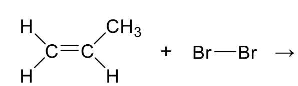
Step 2:

Note:
- product formed is a racemic mixture of 2 enantiomers
- this is because in Step 2, the Br^- nucleophile can attack the sp^2 planar positively charged C from either side of plane, at equal probability, forming a pair of enantiomers with equal proportions

Mechanism

Name:

Step 1:



Step 2:

Step 3:

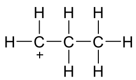
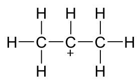
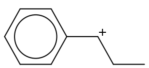
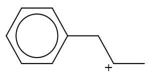
Why electrophilic addition?

➢ Explain why alkenes are reactive towards electrophilic addition

1. Alkenes contain ………………………………. C=C bonds which attract electrophiles
2. The …………….. bonds in C=C are relatively weak and require less energy to break

Comparing relative stability of carbocation

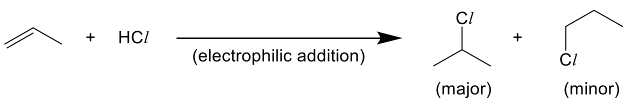
➢ Circle the more stable carbocation in each case

1.  vs 
2.  vs 

Explanation for (2):

- p orbital of positively charged C …………………….. with the π electron cloud of benzene ring
- π electrons can ……………………… onto empty p orbital of positively charged C
- this ……………………… the positive charge on C, stabilising the carbocation

Effect of carbocation stability

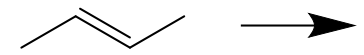


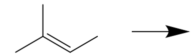
➢ Explain why 2-chloropropane is formed as the major product whereas 1-chloropropane is formed as the minor product.

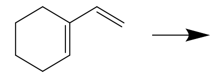
- In the first step, ……………………… and ……………………… can be formed
- the positively charged C in ……………………… is bonded to …………. electron-donating alkyl group, as compared to ……………………… which is bonded to …………. electron-donating alkyl group.
- therefore the positive charge is dispersed to a greater extent in …………………
- hence, …………………… is a more stable carbocation intermediate. It is formed ……………………… and is more available to further react to give the major product.

Deducing products from strong oxidation

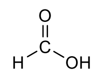
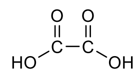
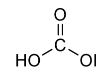
➢ Deduce the products formed when the following compounds are oxidised using $KMnO_4$(aq), H_2SO_4(aq), heat





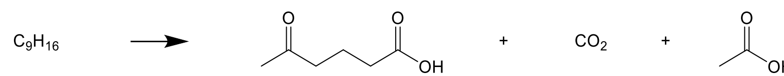


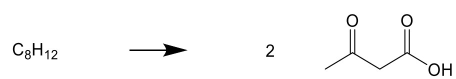
Note:

 &  are able to further oxidise into $CO_2 + H_2O$;  can decompose into $CO_2 + H_2O$

Working backwards from strong oxidation

➢ Deduce possible structures of the alkene which forms the given products upon oxidation with $KMnO_4$(aq), H_2SO_4(aq), heat
You may ignore stereoisomerism

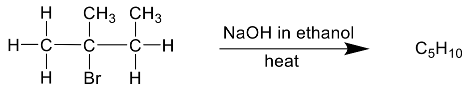




Deducing products from elimination reaction

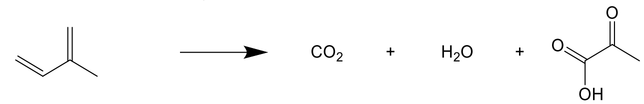
- Draw all possible alkene product formed.
- State the degree of substitution of the alkene formed and hence indicate the major and minor products.

Note: the more substituted alkene is more stable, hence it is the major product



Balancing redox eq$^{\underline{n}}$ with [H] or [O]

- Balance the following equation with [O]

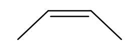
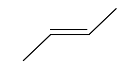


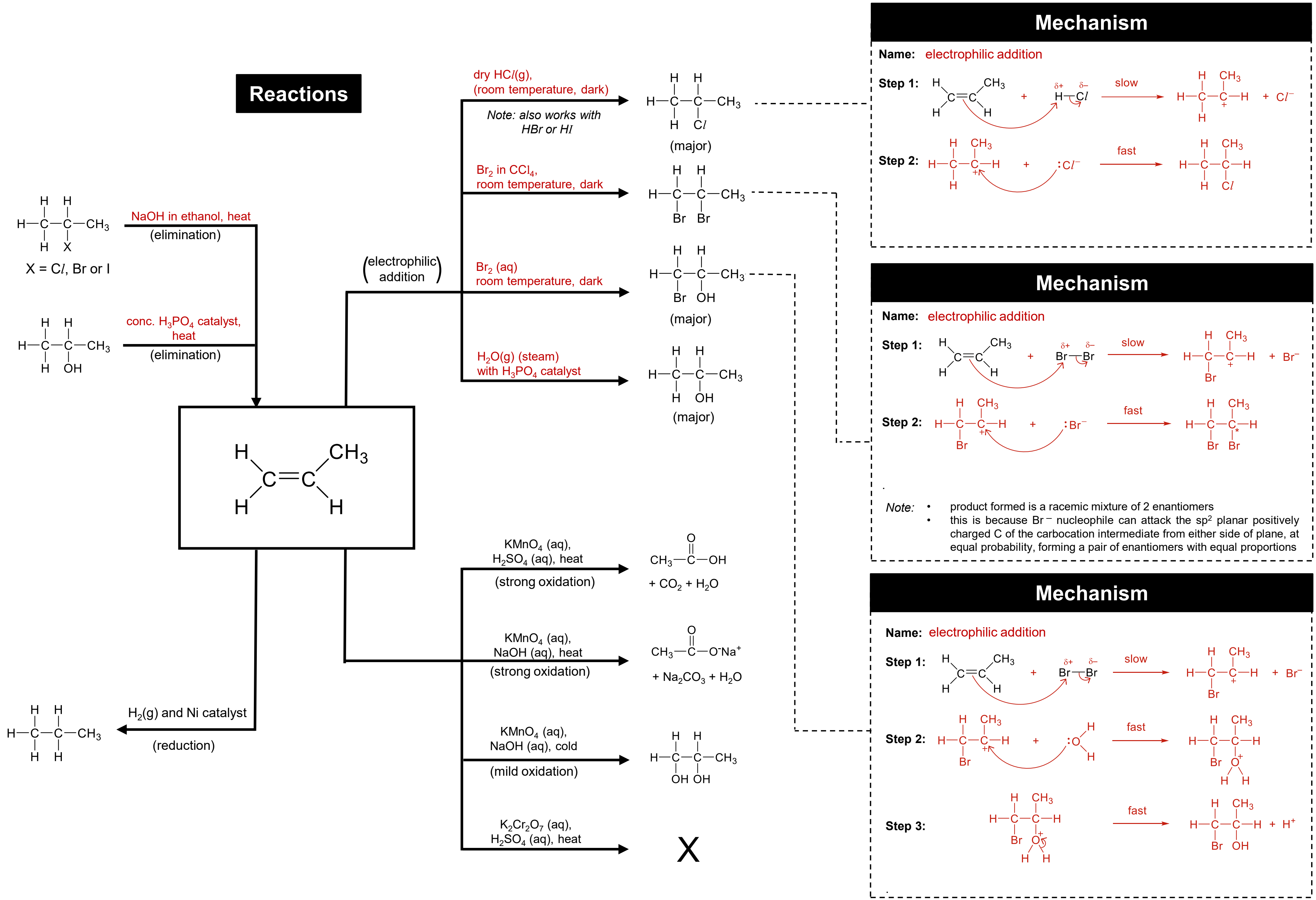
- Balance the following equation with [H]

Method

- balance all atoms (other than H / O)
- balance with H_2O
- then balance H with [H] or O with [O]

Physical properties of cis vs trans

cis vs trans		
melting point	___________ m.p. • packs _____________ efficiently • molecules are further apart • ____________ id-id • ____________ energy required to overcome weaker IMF	___________ m.p. • packs __________ efficiently, • molecules are closer together • _____________ id-id • _____________ energy required to overcome stronger IMF
boiling point	___________ b.p. • ___________ polar • has weak pd-pd interactions • more energy required to overcome stronger IMF	___________ b.p. • ___________ polar • only has id-id interactions • ____________ energy required to overcome weaker IMF



Reactions
NaOH in ethanol, heat
(elimination)
X = Cl, Br or I
conc. H3PO4 catalyst, heat
(elimination)
H2(g) and Ni catalyst
(reduction)
(electrophilic addition)
dry HCl(g), (room temperature, dark)
Note: also works with HBr or HI
(major)
Br2 in CCl4, room temperature, dark
Br2 (aq) room temperature, dark
(major)
H2O(g) (steam) with H3PO4 catalyst
(major)
KMnO4 (aq), H2SO4 (aq), heat
(strong oxidation)
+ CO2 + H2O
KMnO4 (aq), NaOH (aq), heat
(strong oxidation)
+ Na2CO3 + H2O
KMnO4 (aq), NaOH (aq), cold
(mild oxidation)
K2Cr2O7 (aq), H2SO4 (aq), heat
X
Mechanism
Name: electrophilic addition
Step 1:
slow
Step 2:
fast
Mechanism
Name: electrophilic addition
Step 1:
slow
Step 2:
fast
Note:
product formed is a racemic mixture of 2 enantiomers
this is because Br⁻ nucleophile can attack the sp2 planar positively charged C of the carbocation intermediate from either side of plane, at equal probability, forming a pair of enantiomers with equal proportions
Mechanism
Name: electrophilic addition
Step 1:
slow
Step 2:
fast
Step 3:
fast

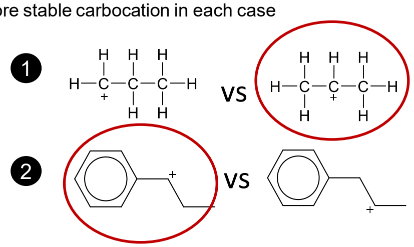
Why electrophilic addition?

➢ Explain why alkenes are reactive towards electrophilic addition

1. Alkenes contain ……electron-rich……… C=C bonds which attract electrophiles
2. The ……π…….. bonds in C=C are relatively weak and require less energy to break

Comparing relative stability of carbocation

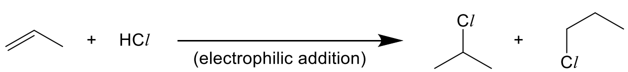
➢ Circle the more stable carbocation in each case



Explanation for (2):

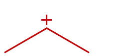
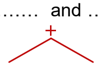
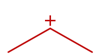
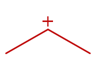
- p orbital of positively charged C ……overlaps……… with the π electron cloud of benzene ring
- π electrons can ……delocalise……… onto empty p orbital of positively charged C
- this ……disperses……… the positive charge on C, stabilising the carbocation

Effect of carbocation stability



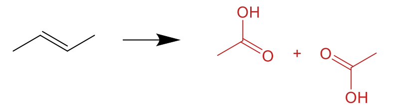
(major) (minor)

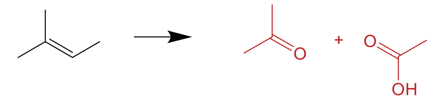
➢ Explain why 2-chloropropane is formed as the major product whereas 1-chloropropane is formed as the minor product.

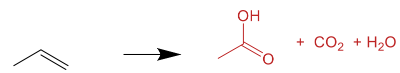
- In the first step,  ……………………… and  ……………………… can be formed
- the positively charged C in  ……………………… is bonded to ……2…… electron-donating alkyl group, as compared to  ……………… which is bonded to ……1…… electron-donating alkyl group.
- therefore the positive charge is dispersed to a greater extent in  …………………
- hence,  ……………… is a more stable carbocation intermediate. It is formed ……more rapidly…… and is more available to further react to give the major product.

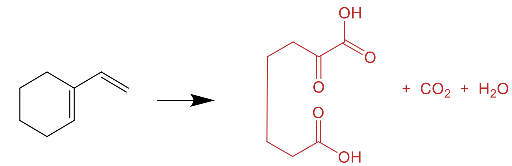
Deducing products from strong oxidation

➢ Deduce the products formed when the following compounds are oxidised using $KMnO_4$(aq), H_2SO_4(aq), heat

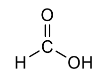
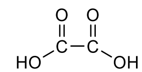
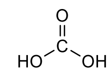








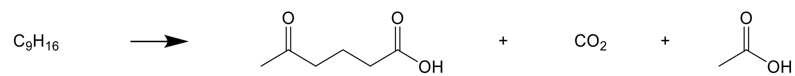
Note:

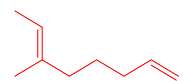
 &  are able to further oxidise into $CO_2 + H_2O$;  can decompose into $CO_2 + H_2O$

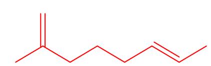
Working backwards from strong oxidation

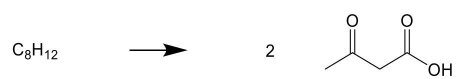
➢ Deduce possible structures of the alkene which forms the given products upon oxidation with $KMnO_4$(aq), H_2SO_4(aq), heat

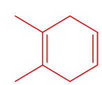
You may ignore stereoisomerism

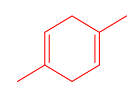








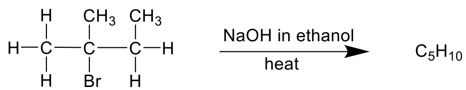


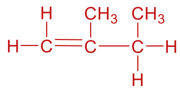


Deducing products from elimination reaction

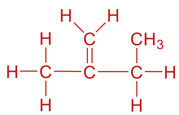
- Draw all possible alkene product formed.
- State the degree of substitution of the alkene formed and hence indicate the major and minor products.

Note: the more substituted alkene is more stable, hence it is the major product

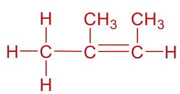




(1°; minor)



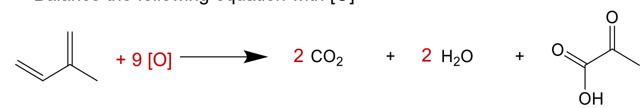
(2° ; minor)



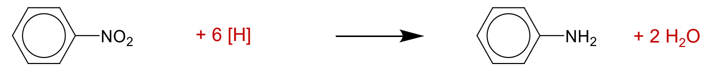
(3° ; major)

Balancing redox eqⁿ with [H] or [O]

- Balance the following equation with [O]



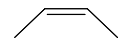
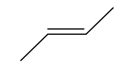
- Balance the following equation with [H]



Method

- balance all atoms (other than H / O)
- balance with H_2O
- then balance H with [H] or O with [O]

Physical properties of cis vs trans

cis vs trans		
melting point	lower m.p. • packs less efficiently • molecules are further apart • weaker id-id • less energy required to overcome weaker IMF	higher m.p. • packs more efficiently, • molecules are closer together • stronger id-id • more energy required to overcome stronger IMF
boiling point	higher b.p. • slightly polar • has weak pd-pd interactions • more energy required to overcome stronger IMF	lower m.p. • non-polar • only has id-id interactions • less energy required to overcome weaker IMF